
GENERAL RELATIVITY TUTORIAL

Problem set 2, solutions, 18.09.2020

■ PROBLEM 5 Constant acceleration, part I.

Consider the equation of motion in SR for a point particle

$$\frac{d\vec{p}}{dt} = \vec{F} , \quad \vec{p} = m\gamma\vec{v} , \quad \gamma = \frac{1}{\sqrt{1-\beta^2}} , \quad \beta = \frac{v}{c} . \quad (5.1)$$

Consider the situation where the force $\vec{F}$ points along the x -axis and has the constant value $F = mg$. Assume that the velocity and position are zero at the time $t = 0$.

(a) Show that the motion of the particle is hyperbolic,

$$\left(x + \frac{c^2}{g}\right)^2 - (x^0)^2 = \left(\frac{c^2}{g}\right)^2 , \quad (5.2)$$

where $x^0 = ct$ and $y = z = 0$.

(b) Let $\mathbf{I}$ denote the inertial system where the particle is at rest at $t = 0$. Show that the proper time of a clock following the accelerated particle is given by

$$\tau = \frac{c}{g} \operatorname{arcsinh} \left(\frac{gx^0}{c^2} \right) , \quad (5.3)$$

i.e.

$$x^0 = \frac{c^2}{g} \sinh \left(\frac{g\tau}{c} \right) . \quad (5.4)$$

(c) Show that the transformation from an inertial system $\mathbf{I}'$, where the accelerated particle is at rest at proper time τ at $x' = 0 = x'^0$ (and $y' = z' = 0$), to $\mathbf{I}$ is given by

$$x = \frac{c^2}{g} \left[\cosh \left(\frac{g\tau}{c} \right) - 1 \right] + x' \cosh \left(\frac{g\tau}{c} \right) + x'^0 \sinh \left(\frac{g\tau}{c} \right) , \quad (5.5)$$

$$x^0 = \frac{c^2}{g} \sinh \left(\frac{g\tau}{c} \right) + x' \sinh \left(\frac{g\tau}{c} \right) + x'^0 \cosh \left(\frac{g\tau}{c} \right) . \quad (5.6)$$

SOLUTION: (a) The differential equation to solve (in the nontrivial x direction) is, $(d/dt)[m\gamma v] = mg$. Integrating once gives, $u \equiv \gamma v = gt = v/\sqrt{1-(v/c)^2}$, from which one obtains,

$$v(t)dt = \frac{dx}{dt}dt = \frac{gtdt}{\sqrt{1+(gt/c)^2}} . \quad (5.7)$$

Integrating this gives, $x(t) = \frac{c^2}{g} \left[\sqrt{1+(gt/c)^2} - 1 \right]$, which is equivalent to (5.2).

(b) Proper time τ is defined as,

$$d\tau^2 = dt^2 - \frac{dx^2}{c^2} = dt^2 \left(1 - \frac{v^2}{c^2} \right) = \frac{dt^2}{\gamma^2} . \quad (5.8)$$

Along the geodesic we have, $dx/dx^0 = dx/(cdt) = x^0/\sqrt{(x^0)^2 + (c^2/g)^2}$, which can be obtained by differentiating (5.2). Upon inserting this into (5.8) one obtains (5.3),

$$\tau = \frac{c}{g} \int_0^{x^0} \frac{d\tilde{x}^0}{\sqrt{(\tilde{x}^0)^2 + (c^2/g)^2}} = \frac{c}{g} \operatorname{arcsinh} \left(\frac{gx^0}{c^2} \right) . \quad (5.9)$$

Inverting this one obtains, $x^0(\tau) = \frac{c^2}{g} \sinh \left(\frac{g\tau}{c} \right)$, which equals to (5.4).

(c) Lorentz transformations are given by,

$$\begin{aligned} x &= \bar{x}(\tau) + \gamma(\tau) \left[x' + \frac{v(\tau)}{c} x'^0 \right] \\ x^0 &= \bar{x}^0(\tau) + \gamma(\tau) \left[x'^0 + \frac{v(\tau)}{c} x' \right] , \end{aligned} \quad (5.10)$$

where $\bar{x}(\tau)$ and $\bar{x}^0(\tau)$ account for the noninertial nature of the coordinate transformation and have been calculated in parts (a) and (b) of the problem, $\bar{x}(\tau) = \frac{c^2}{g} [\cosh \left(\frac{g\tau}{c} \right) - 1]$, $\bar{x}^0(\tau) = \frac{c^2}{g} \sinh \left(\frac{g\tau}{c} \right)$.

To complete the task, we ought to find the functional dependence on proper time of, $v(\tau)$, $u(\tau) = \gamma v(\tau)$ and $\gamma(\tau) = u(\tau)/v(\tau)$. Firstly, note that

$$u(\tau) = gt(\tau) = \frac{gx^0(\tau)}{c} = c \sinh \left(\frac{g\tau}{c} \right) = c \sqrt{\gamma^2 - 1} ,$$

from which it follows, $\gamma(\tau) = \cosh(g\tau/c)$. Inserting these results into (5.10) gives (5.6).

■ PROBLEM 6 Rotating coordinate system.

In order to find the line element of a uniformly rotating reference frame, we can start from flat space in cylindrical coordinates,

$$ds'^2 = g'_{\mu\nu} dx'^\mu dx'^\nu = -c^2 dt'^2 + d\rho'^2 + dz'^2 + \rho'^2 d\varphi'^2 , \quad (6.1)$$

and perform the following spatial transformation for axis of rotation z ,

$$\rho = \rho' , \quad z = z' , \quad \varphi = \varphi' + \omega t , \quad (6.2)$$

where ω is the constant angular velocity of rotation.

(a) What is the line element ds^2 of the rotating coordinate system?

(b) What would be the circumference of a circle in terms of the coordinate ρ as measured by an observer in a rotating frame? Discuss the physical meaning of the case when $\rho \geq \frac{c}{\omega}$.

Hint: An observer on a given space-time measures the time interval between two events as $d\tau^2 = -g_{00}dt^2$, and the space interval as $dl^2 = \left(g_{ij} - \frac{g_{0i}g_{0j}}{g_{00}}\right) dx^i dx^j$.

SOLUTION: (a) Making use of, $d\rho' = d\rho$, $dz' = dz$, $d\varphi' = d\varphi - \omega dt$. Supplying this with $dt' = dt$ (this last relation says that we are measuring time in the rotating frame in units of time in the lab frame, which is not the time measured in the rotating frame), one obtains the line element,

$$ds^2 = -(c^2 - \omega^2 \rho^2)dt^2 + d\rho^2 + dz^2 + \rho^2 d\varphi^2 - 2\omega \rho^2 dt d\varphi. \quad (6.3)$$

(b) The time as measured in the rotating frame is also the proper time of an observer at rest in that frame, *i.e.* $d\tau = \sqrt{1 - (\omega\rho/c)^2}dt$, which means that – if $\rho = \text{constant}$, then $\Delta\tau = \sqrt{1 - (\omega\rho/c)^2}\Delta t$, *i.e.* the observer in the rotating frame experiences a time dilatation. This should not surprise us, as that is an accelerating observer. Because the metric (6.3) is not diagonal, the proper length measured by the rotating observer

$$d\ell^2 = d\rho^2 + dz^2 + \rho^2 \left(1 + \frac{\omega^2 \rho^2}{c^2 - \omega^2 \rho^2}\right) d\varphi^2 = d\rho^2 + dz^2 + \frac{\rho^2}{1 - (\omega\rho/c)^2} d\varphi^2 \quad (6.4)$$

This means that there is a critical radial distance, $\rho_{\text{cr}} = c/\omega$, at which the proper distance becomes infinite. Beyond that distance, $\rho > \rho_{\text{cr}}$, the angular coordinate becomes time-like (the coefficient of the metric becomes negative), which is similar to what happens at the event horizon of a black hole for the radial coordinate. This means that the spatial metric (6.4) only makes sense for distances $\rho < c/\omega$. To get a better grasp of what it represents, let us transform it into a new radial coordinate, $r = \rho/\sqrt{1 - (\omega\rho/c)^2}$, in terms of which the circumference of a circle at radius r is $2\pi r$. Using this radial coordinate, the infinitesimal proper length (6.4) becomes,

$$d\ell^2 = \frac{dr^2}{[1 + (\omega r/c)^2]^3} + dz^2 + r^2 d\varphi^2, \quad (6.5)$$

and the range of this coordinate is $0 \leq r < \infty$. From this we conclude that – as r increases – the proper radial distance exhibits a contraction, $dr/[1 + (\omega r/c)^2]^{3/2}$, which at large radial distances (close to ρ_{cr}) behaves as, $\sim c^3 dr/(\omega r)^3$, *i.e.* it tends to zero. This means that the metric corresponds to a space-time with an infinite extent in the z direction, but a finite extent in the radial direction, *i.e.* the space time has geometry of a cylinder. One physical implication of this analysis is that one cannot rotate a disk of radius R with a speed that is equal or greater than $\omega_{\text{cr}} = c/R$. The reason is simple to understand. A piece of the disk of mass dm at the rim of the rotating disk move with velocity $v = \omega R$, experiences an increase in energy according to, $\delta E = (\delta m)\gamma c^2 = (\delta m)c^2/\sqrt{1 - (\omega R/c)^2}$, which diverges as ω reaches the critical value, providing the physical explanation for why an angular velocity greater or equal to the critical is impossible to reach.